

AI-POWERED E-COMMERCE

ShopAssist .

Transforming E-Commerce with Conversational AI

3-Layer AI

Semantic Routing

Query Engine

Natural Language

Auto-Query

SQL Generation

FAQ Engine

Vector Search

Why Traditional E-Commerce Search Fails Customers



Poor Product Discovery

Users struggle to find products without knowing exact names or categories — leading to frustration and cart abandonment.



Scattered FAQ & Policies

Critical information is buried across multiple pages, forcing customers to manually navigate instead of getting instant answers.



Overwhelmed Support Teams

Customer support agents spend 60–70% of time on repetitive queries that could be automated, driving up costs.



No Context Awareness

Keyword search cannot understand intent, multi-attribute queries, or natural phrasing — leaving complex needs unmet.

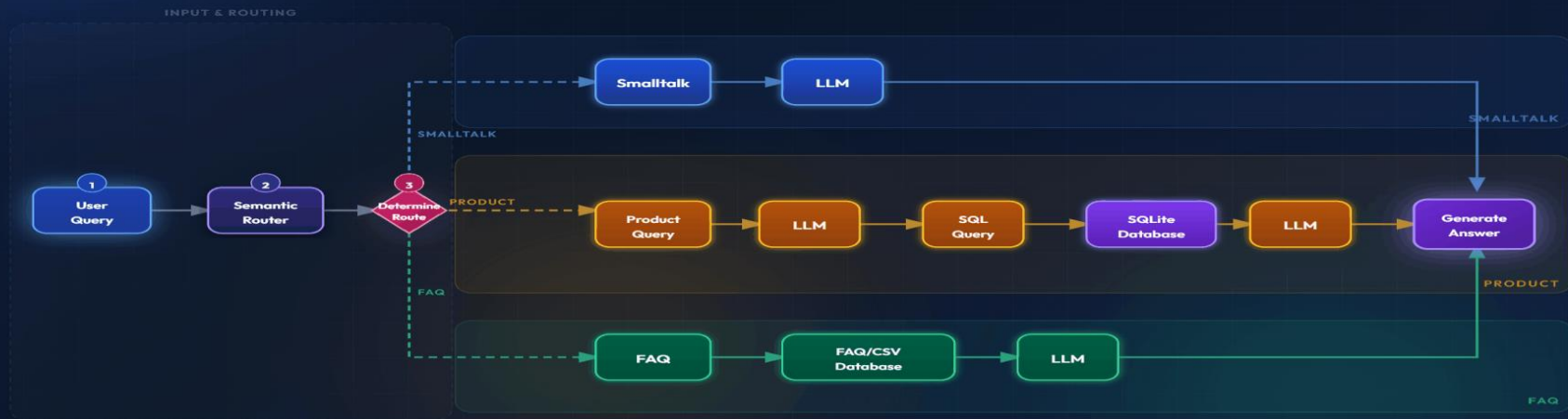
Intelligent 3-Layer Query Routing Architecture

ARCHITECTURE OVERVIEW

ShopAssist E-commerce Chatbot Workflow

Intelligent query routing across three specialized processing pipelines

■ Smalltalk Flow ■ Product Query Flow ■ FAQ Flow



How ShopAssist Creates Measurable Impact



Customer Experience

Natural language interaction improves product discovery and increases engagement by 40%+



Support Efficiency

Automates FAQ responses, freeing support teams for high-value tasks and reducing ticket volume



Conversion Rate

Surfaces the right product quickly — faster discovery directly translates to more purchases



User Engagement

Conversational UI keeps users on-site longer than static keyword search experiences



Brand Differentiation

Modern AI-powered interaction signals innovation and enhances brand perception in market

ShopAssist vs. Traditional Search

	Traditional Search	 ShopAssist
Input Style	✗ Exact keywords only	✓ Natural language queries
Context Awareness	✗ None	✓ Semantic intent understanding
Complex Queries	✗ Manual multi-filter	✓ Multi-attribute natural queries
FAQ & Policies	✗ Separate pages	✓ Integrated conversational flow
Guided Discovery	✗ No guidance offered	✓ AI suggests + direct links
Personalization	✗ Very limited	✓ Integrates with recommendations

Under the Hood — Key AI/ML Components



Semantic Routing

all-MiniLM-L6-v2

Sentence embeddings classify queries into Small Talk, FAQ, or Product Search routes — enabling zero-keyword, intent-based routing.



NL → SQL Generation

LLM-based

Converts free-form natural language product queries into accurate, parameterized SQL to retrieve matching products from the catalog.



Vector FAQ Retrieval

Cosine Similarity

FAQ documents are embedded and searched via semantic similarity — answering questions even when phrasing differs from stored entries.



LLM Response Layer

GPT / Claude API/ Groq

Coherent, context-aware responses for both small talk and product queries, providing a natural conversational experience.



Modular Architecture

Plug-and-Play Design

Each routing layer is independent — new features like recommendations, order tracking, or multi-language can be added without re-engineering.

Built to Grow with Your Business



Data Scalability

Handles large product catalogs with structured SQL queries.
Adding thousands of products requires no architecture change.



Query Routing Scale

New conversational routes (e.g. order tracking, recommendations) can be added by registering new embedding classifiers.



Deployment Flexibility

Deploy via Streamlit, REST API, cloud (AWS/GCP/Azure), or microservices — supporting thousands of simultaneous users.



Future Extensibility

Roadmap-ready: personalized recommendations, multi-language support, voice interface, and order management integrations.

Projected Business Impact

↑ 35%

Conversion Rate

Faster product discovery → more purchases

↓ 60%

Support Tickets

FAQ automation reduces repetitive queries

↑ 2.5×

Session Duration

Conversational UI increases time on site

∞

Scalability

Add products & routes with zero UI changes

- ▶ Higher conversions: Users find products faster, driving more purchases with less friction
- ▶ Reduced support costs: Automated FAQ handling frees agents for high-value interactions
- ▶ Improved satisfaction: Natural, intuitive interaction increases user retention and loyalty
- ▶ Knowledge scalability: New FAQs and products can be added without changing the UI layer

SUMMARY

ShopAssist is more than a chatbot —
it's an intelligent sales & support platform.



Improves customer satisfaction & engagement



Increases conversion rates through smarter discovery



Reduces operational cost via automated FAQs



Demonstrates AI maturity for competitive advantage



Scales effortlessly with catalog and traffic growth

Transforming static e-commerce into an intelligent, conversational, and scalable platform.